

Use mental multiplication and division strategies

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Calculate $25 \times 3,600$ mentally. Copy or complete the first step.

2. Calculate $4,800 \div 16$ mentally.

3. Calculate 125×64 using factors.

4. Miro says: Miro divides by 4 when multiplying by 25 but forgets to multiply by 100. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate $25 \times 3,600$ mentally. Copy or complete the first step.
Answer 90,000.
2. Calculate $4,800 \div 16$ mentally.
Answer 300.
3. Calculate 125×64 using factors.
Answer 8,000.
4. Miro says: Miro divides by 4 when multiplying by 25 but forgets to multiply by 100. Is this correct? Explain.
Answer Since $25 = 100 \div 4$, divide by 4 and then multiply by 100.

Use mental multiplication and division strategies

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Calculate $4,800 \div 16$ mentally.

2. Calculate 125×64 using factors.

3. Find three efficient mental methods for 48×250 .

4. Identify and correct this misconception: Miro divides by 4 when multiplying by 25 but forgets to multiply by 100.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. Calculate $4,800 \div 16$ mentally.
Answer 300.
2. Calculate 125×64 using factors.
Answer 8,000.
3. Find three efficient mental methods for 48×250 .
Answer 12,000. Methods may use $250 = 1,000 \div 4$ or $48 \div 4 \times 1,000$.
4. Identify and correct this misconception: Miro divides by 4 when multiplying by 25 but forgets to multiply by 100.
Answer Since $25 = 100 \div 4$, divide by 4 and then multiply by 100.
5. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Use mental multiplication and division strategies

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Find three efficient mental methods for 48×250 .

2. Explain why this reasoning fails: Miro divides by 4 when multiplying by 25 but forgets to multiply by 100.

3. Create a new example that tests the same idea as: Calculate 125×64 using factors.

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Find three efficient mental methods for 48×250 .
Answer 12,000. Methods may use $250 = 1,000 / 4$ or $48 / 4 \times 1,000$.
2. Explain why this reasoning fails: Miro divides by 4 when multiplying by 25 but forgets to multiply by 100.
Answer Since $25 = 100 / 4$, divide by 4 and then multiply by 100.
3. Create a new example that tests the same idea as: Calculate 125×64 using factors.
Answer Answers vary. The example and solution must preserve the mathematical structure.
4. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
5. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.